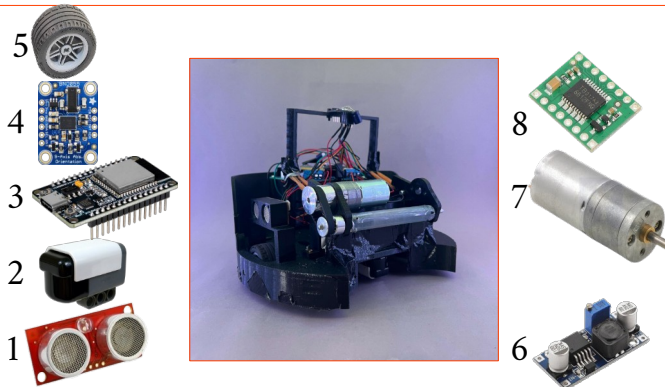


WHO WE ARE ?

We are the **NO NAME PENGUINS**, a German academic high school team competing in the Soccer Entry category. Our team consists of **Luna** (15), who is responsible for the software, and **Emily** (16), who develops the hardware. This year marks our second time qualifying for the European Championship. Last year, we won the championship using a very old NXT LEGO brick. This year, we are upgrading our robot and have significantly improved our hardware.



Component	Model	Number
1 UltraSonic	(srf08)	4
2 IR-Seeker	(from HiTechnic)	1
3 Microcontroller	(Esp32)	1
4 Motor driver	(Pololu md08a0J7303)	2
5 Wheels	(Lego)	2
6 Voltage Converter	(LM2596 DC-DC HW-411)	1
7 Motor	(Pololu MP 12V 9.7:1)	3
8 Compass	(BN0055)	1

ROBOT STRUCTURE

In general, we program our robot using C++. Our robot is equipped with a dribbler that can catch the ball and hold it securely within the capturing zone, allowing for better control during the game. Additionally, we use a custom-built platform with various sensors that help the robot detect its surroundings and react accordingly. The construction of our robot combines LEGO components with mostly 3D-printed parts, enabling us to create a flexible and optimized design.